

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. The following sequences can be shown to be telescoping. Find s_n for each series and use this to determine the sum of the series.

a)
$$\sum_{k=1}^{\infty} \left[\cos\left(\frac{1}{k}\right) - \cos\left(\frac{1}{k+1}\right) \right]$$

We take the sum of the first n terms of the series.

$$\begin{aligned} s_n &= \left[\cos\left(\frac{1}{1}\right) - \cos\left(\frac{1}{2}\right) \right] + \left[\cos\left(\frac{1}{2}\right) - \cos\left(\frac{1}{3}\right) \right] + \left[\cos\left(\frac{1}{3}\right) - \cos\left(\frac{1}{4}\right) \right] + \dots + \left[\cos\left(\frac{1}{n}\right) - \cos\left(\frac{1}{n+1}\right) \right] \\ &= \cos(1) - \cos\left(\frac{1}{n+1}\right) \end{aligned}$$

Now, to determine if the series converges, we must take the limit of s_n as $n \rightarrow \infty$.

$$\text{We see that } \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left[\cos(1) - \cos\left(\frac{1}{n+1}\right) \right] = \cos(1) - \cos(0) = \cos(1) - 1.$$

$$\text{Therefore, the series converges, and we say that } \sum_{k=1}^{\infty} \left[\cos\left(\frac{1}{k}\right) - \cos\left(\frac{1}{k+1}\right) \right] = \cos(1) - 1.$$

b)
$$\sum_{k=1}^{\infty} \frac{3}{k^2 + 3k}$$

Since we don't have a subtraction of two terms here we will first attempt to write the formula $\frac{3}{k^2 + 3k}$.

$$\frac{3}{k^2 + 3k} = \frac{3}{k(k+3)} = \frac{A}{k} + \frac{B}{k+3} = \frac{A(k+3) + Bk}{k(k+3)}$$

This implies that $3 = A(k+3) + Bk$.

$$\text{If } k = 0 \text{ then } 3 = 3A \rightarrow A = 1$$

$$\text{If } k = -3 \text{ then } 3 = -3B \rightarrow B = -1$$

$$\text{Therefore, } \sum_{k=1}^{\infty} \frac{3}{k^2 + 3k} = \sum_{k=1}^{\infty} \left[\frac{1}{k} - \frac{1}{k+3} \right]$$

We now determine the sum of the first n terms of the series.

$$\begin{aligned} s_n &= \left[\frac{1}{1} - \frac{1}{4} \right] + \left[\frac{1}{2} - \frac{1}{5} \right] + \left[\frac{1}{3} - \frac{1}{6} \right] + \left[\frac{1}{4} - \frac{1}{7} \right] + \left[\frac{1}{5} - \frac{1}{8} \right] + \dots + \left[\frac{1}{n-3} - \frac{1}{n} \right] + \left[\frac{1}{n-2} - \frac{1}{n+1} \right] + \left[\frac{1}{n-1} - \frac{1}{n+2} \right] + \left[\frac{1}{n} - \frac{1}{n+3} \right] \\ &= 1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \end{aligned}$$

Now, to determine if the series converges, we must take the limit of s_n as $n \rightarrow \infty$.

$$\text{We see that } \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left[1 + \frac{1}{2} + \frac{1}{3} - \frac{1}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \right] = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}.$$

Therefore, the series converges, and we say that $\sum_{k=1}^{\infty} \frac{3}{k^2 + 3k} = \frac{11}{6}$.

2. Determine if the following series are geometric or not. If it is geometric, then find a & r and determine if the series converges. If it converges, find its sum.

a) $\sum_{n=1}^{\infty} 2(3)^n$

We see that $\sum_{n=1}^{\infty} 2(3)^n = \sum_{n=1}^{\infty} 6(3)^{n-1}$. This is a geometric series with $a = 6$ and $r = 3$. Since $|r| > 1$ the series diverges.

b) $\sum_{n=0}^{\infty} 2(3)^n$

We see that $\sum_{n=0}^{\infty} 2(3)^n = \sum_{n=0}^{\infty} 2(3)^n$. This is a geometric series with $a = 2$ and $r = 3$. Since $|r| > 1$ the series diverges.

c) $\sum_{n=1}^{\infty} n \left(-\frac{1}{2} \right)^n$

This is NOT a geometric series since it cannot be rewritten to be in the form $\sum_{n=1}^{\infty} a(r)^{n-1}$ since the variable n is not exclusively in the power.

d) $\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}}$

We see that $\sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}} = \sum_{n=1}^{\infty} e \frac{e^{n-1}}{3^{n-1}} = \sum_{n=1}^{\infty} e \left(\frac{e}{3} \right)^{n-1}$. This is a geometric series with $a = e$ and $r = \frac{e}{3}$. Since $|r| < 1$ the series converges.

$$\text{Therefore, } \sum_{n=1}^{\infty} \frac{e^n}{3^{n-1}} = \frac{e}{\left(1 - \frac{e}{3} \right)} = \frac{e}{\left(\frac{3-e}{3} \right)} = \frac{3e}{3-e}$$

3. Determine the sum of the following series $\sum_{n=0}^{\infty} (2^{n+1} - 1) \left(\frac{1}{9}\right)^n$.

Note that

$$\begin{aligned} \sum_{n=0}^{\infty} (2^{n+1} - 1) \left(\frac{1}{9}\right)^n &= \sum_{n=0}^{\infty} \left(2^{n+1} \left(\frac{1}{9}\right)^n - \left(\frac{1}{9}\right)^n \right) = \sum_{n=0}^{\infty} \left(2 \cdot 2^n \left(\frac{1}{9}\right)^n - \left(\frac{1}{9}\right)^n \right) \\ &= \sum_{n=0}^{\infty} \left(2 \left(\frac{2}{9}\right)^n - \left(\frac{1}{9}\right)^n \right) = \sum_{n=0}^{\infty} 2 \left(\frac{2}{9}\right)^n - \sum_{n=0}^{\infty} \left(\frac{1}{9}\right)^n \\ &= \frac{2}{1 - 2/9} - \frac{1}{1 - 1/9} \\ &= \frac{2}{7/9} - \frac{1}{8/9} = \frac{18}{7} - \frac{9}{8} = \frac{144}{56} - \frac{63}{56} \\ &= \frac{81}{56} \end{aligned}$$

4. Create an infinite series which converges to the value $\frac{\pi}{2}$.

We will create a geometric series that sums to this value.

Note that $\frac{\pi}{2} = \frac{1}{(2/\pi)} = \frac{1}{1 - \left(1 - \frac{2}{\pi}\right)} = \frac{1}{1 - \left(\frac{\pi - 2}{\pi}\right)}$. This resembles the sum of a geometric series with

$a = 1$ and $r = \frac{\pi - 2}{\pi}$ and clearly $|r| < 1$. Therefore, the series $\sum_{n=1}^{\infty} 1 \left(\frac{\pi - 2}{\pi}\right)^{n-1}$ converges to $\frac{\pi}{2}$.

OR

We will create a series that converges to 1 and then multiply the series by $\frac{\pi}{2}$.

Note that $1 = \frac{1/2}{1 - (1/2)}$. This resembles the sum of a geometric series with $a = \frac{1}{2}$ and $r = \frac{1}{2}$ and clearly

$|r| < 1$. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{2}\right)^{n-1}$ converges to 1 and thus $\frac{\pi}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} = \sum_{n=1}^{\infty} \frac{\pi}{4} \left(\frac{1}{2}\right)^{n-1}$ must

converge to $\frac{\pi}{2}$.

5. When I was a teenager, a friend and I stole soda from my parents. I took the soda bottle, drank half of what's there, and then passed it to my friend. My friend drank half of what was left, and passed it back to me. I again drank half of what was left, and passed it back to him. We kept up with this until the entire soda bottle was empty. Approximately what percentage of the soda did I drink?

We note that the amount of soda drank is represented by the infinite sum $\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} + \dots$.

We can rewrite this series in summation notation as $\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \frac{1}{128} + \dots = \sum_{n=1}^{\infty} \frac{1}{2(4^{n-1})}$.

Then we see that $\sum_{n=1}^{\infty} \frac{1}{2(4^{n-1})} = \sum_{n=1}^{\infty} \frac{4}{2 \cdot 4^n} = \sum_{n=1}^{\infty} \frac{2}{4^n} = \sum_{n=1}^{\infty} \frac{2}{4} \left(\frac{1}{4}\right)^{n-1} = \frac{1/2}{1-1/4} = \frac{1/2}{3/4} = \frac{4}{6} = \frac{2}{3} \approx 67\%$

So, I drank about 67% of the soda.

6. Later in this course we will show that $\sum_{n=0}^{\infty} \frac{1}{n!} = e$. Use this fact to show that for every $a > 2$ it must be the case that $\sum_{n=0}^{\infty} \frac{1}{a^n} < \sum_{n=0}^{\infty} \frac{1}{n!}$.

We see that $\sum_{n=0}^{\infty} \frac{1}{a^n} = \sum_{n=0}^{\infty} \left(\frac{1}{a}\right)^n$ is a geometric series and if $a > 2$ then $|r| < 1$ and this series converges.

Specifically, $\sum_{n=0}^{\infty} \left(\frac{1}{a}\right)^n = \frac{1}{1 - \left(\frac{1}{a}\right)} = \frac{1}{\left(\frac{a-1}{a}\right)} = \frac{a}{a-1} = \frac{a-1+1}{a-1} = 1 + \frac{1}{a-1}$.

Since $a > 2$ it must be the case that $1 + \frac{1}{a-1} < 2 < e = \sum_{n=0}^{\infty} \frac{1}{n!}$ series sums to a value less than 2.

7. Consider the following expression

$$\sqrt{2\sqrt{4\sqrt{8\sqrt{16}\dots}}}$$

a) Find a way to rewrite this expression so it can be expressed as 2^x , where x represents some expression.

$$\text{Note that } \sqrt{2\sqrt{4\sqrt{8\sqrt{16}\dots}}} = 2^{1/2} \times 2^{2/4} \times 2^{3/8} \times 2^{4/16} \times 2^{5/32} \times \dots = 2^{\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \dots}$$

$$\text{Here } x = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \dots$$

b) Take your value of x from part (a) and subtract from it $\frac{x}{2}$. Use this result to help determine the exact value of x and the sum.

We calculate

$$x = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \dots$$

$$\frac{x}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots$$

$$\frac{x}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots$$

So, this means that

$$\frac{x}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots \rightarrow x = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\rightarrow x = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} = \frac{1}{1 - \frac{1}{2}} = 2$$

c) Use your answer from part (b) to establish the value of the given expression

$$\text{Therefore, } \sqrt{2\sqrt{4\sqrt{8\sqrt{16}\dots}}} = 2^{\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \dots} = 2^2 = 4$$

8. If m is a positive integer such that $\sum_{n=1}^{\infty} \log \left(1 - \frac{1}{(n+5)^2} \right) = \log \left(\frac{m}{m+1} \right)$, then what is the value of m ?

(Hint: Start by trying to algebraically rewrite the expression inside the logarithm and see if you can rewrite the series as a telescoping series)

$$\sum_{n=1}^{\infty} \log \left(1 - \frac{1}{(n+5)^2} \right) = \sum_{n=1}^{\infty} \log \left(\left(1 - \frac{1}{(n+5)} \right) \left(1 + \frac{1}{(n+5)} \right) \right) = \sum_{n=1}^{\infty} \log \left(\frac{n+4}{n+5} \cdot \frac{n+6}{n+5} \right).$$

We see that

$$s_k = \log \left(\frac{5}{6} \cdot \frac{7}{6} \right) + \log \left(\frac{6}{7} \cdot \frac{8}{7} \right) + \log \left(\frac{7}{8} \cdot \frac{9}{8} \right) + \dots + \log \left(\frac{k+4}{k+5} \cdot \frac{k+6}{k+5} \right) = \log \left(\frac{5}{6} \cdot \frac{k+6}{k+5} \right)$$

We calculate that $\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} \log \left(\frac{5}{6} \cdot \frac{k+6}{k+5} \right) = \log \left(\frac{5}{6} \right)$.

By definition we then know that $\sum_{n=1}^{\infty} \log \left(1 - \frac{1}{(n+5)^2} \right) = \lim_{k \rightarrow \infty} s_k = \log \left(\frac{5}{6} \right)$ and therefore $m = 5$.